Name_____

Date_____

AP Chemistry
Chapters 6 & 7 Test
Electronic Structure & Periodic Properties
Practice Test B

Section I: Multiple Choice, 40 questions, written to be done without a calculator. Choose the best answer and write its letter in the blank.

Section II: Free Response, 2 questions (20 points). Calculator allowed. For full credit, write the formula, substitute with units, and give the answer with units and correct significant figures. Answer "explain" parts in complete sentences.

A periodic table is on the page after the test. Constants are not provided.

Section I: Multiple Choice

Directions: Each question is followed by four answer choices. Select the one that is best in each case.

_____ **1.** Which lists these types of radiation in order of DECREASING frequency?

(A) radio > infrared > visible > gamma rays
(B) gamma rays > visible > infrared > radio
(C) visible > gamma rays > radio > infrared
(D) gamma rays > radio > visible > infrared

_____ **2.** What is the frequency of light with a wavelength of 750 nm?

(A) $2.50\times10^{-15}\ s^{-1}$
(B) $2.25\times10^{2}\ s^{-1}$
(C) $4.00\times10^{5}\ s^{-1}$
(D) $4.00\times10^{14}\ s^{-1}$

_____ **3.** An FM radio station broadcasts at 100 MHz. What is the wavelength of these radio waves?

(A) 3.00 m
(B) 3.00×10^{3} m
(C) 3.00×10^{6} m
(D) 3.00×10^{16} m

_____ **4.** Which expression gives the wavelength, in nanometers, of a photon with an energy of 4.0×10^{-19} J?

(A) $(6.626\times10^{-34})(4.0\times10^{-19}) / (2.998\times10^{8}) \times 10^{9}$
(B) $(6.626\times10^{-34})(2.998\times10^{8}) / (4.0\times10^{-19}) \times 10^{-9}$
(C) $(4.0\times10^{-19}) / [(6.626\times10^{-34})(2.998\times10^{8})] \times 10^{9}$
(D) $(6.626\times10^{-34})(2.998\times10^{8}) / (4.0\times10^{-19}) \times 10^{9}$

_____ **5.** Light of wavelength λ has a photon energy E. If the wavelength is tripled to 3λ, what is the new photon energy?

(A) 3E
(B) E/3
(C) E/9
(D) E (unchanged)

_____ **6.** About how much energy is carried by 1 mol of photons of 200 nm ultraviolet light?

(A) 9.9×10^{-19} kJ/mol

(B) 6.0×10^{-7} kJ/mol

(C) 6.0×10^{2} kJ/mol

(D) 6.0×10^{5} kJ/mol

_____ **7.** Four lines in the hydrogen emission spectrum come from electrons falling to n = 1 from n = 2, 3, 4 and 5. Their wavelengths are 94.9 nm, 97.2 nm, 102.5 nm and 121.5 nm. Which wavelength belongs to the n = 3 → n = 1 transition?

(A) 94.9 nm

(B) 97.2 nm

(C) 102.5 nm

(D) 121.5 nm

_____ **8.** Red light shines on a metal surface, and no electrons are ejected. Which change is most likely to cause electrons to be ejected?

(A) Make the red light brighter

(B) Shine the red light on the metal for a longer time

(C) Switch to infrared light

(D) Switch to ultraviolet light

_____ **9.** A photon with an energy of 5.0×10^{-19} J strikes a metal. The minimum energy needed to remove an electron from this metal is 3.2×10^{-19} J. What is the kinetic energy of the ejected electron?

(A) 1.8×10^{-19} J

(B) 5.0×10^{-19} J

(C) 8.2×10^{-19} J

(D) No electron is ejected.

_____ **10.** When light from excited hydrogen gas passes through a prism, only a few bright lines at specific wavelengths appear instead of a continuous rainbow. Which best explains this?

(A) Hydrogen atoms give off only visible light, so the other colors are missing.

(B) The electron in hydrogen can have only certain energies, so only photons whose energies equal the gaps between levels are emitted.

(C) The prism absorbs most of the wavelengths that pass through it.

(D) Electrons in hydrogen can have any energy, but they give off light only when they reach n = 1.

_____ **11.** An electron in a hydrogen atom moves from n = 6 to n = 4. Which describes ΔE for the electron and the photon involved?

(A) $\Delta E > 0$; a photon is absorbed.

(B) $\Delta E > 0$; a photon is emitted.

(C) $\Delta E < 0$; a photon is absorbed.

(D) $\Delta E < 0$; a photon is emitted.

_____ **12.** Which statement about the energy levels in the Bohr model of hydrogen is correct?

(A) As n increases, the energy becomes less negative and the levels get closer together.

(B) As n increases, the energy becomes more negative and the levels spread farther apart.

(C) The levels are equally spaced in energy.

(D) The n = 1 level has the highest (least negative) energy.

_____ **13.** What is the energy of an electron in the n = 3 level of a hydrogen atom?

(A) -1.96×10^{-17} J

(B) -7.27×10^{-19} J

(C) -2.42×10^{-19} J

(D) $+2.42\times10^{-19}$ J

_____ **14.** Which transition in a hydrogen atom EMITS the photon with the highest frequency?

(A) n = 1 → n = 5

(B) n = 6 → n = 2

(C) n = 2 → n = 1

(D) n = 4 → n = 1

_____ **15.** Which expression gives the wavelength, in meters, of the photon emitted when an electron in a hydrogen atom falls from n = 6 to n = 1? (h = Planck's constant, c = speed of light)

(A) hc / [2.18×10^{-18} J (1 − 1/6)]

(B) hc / [2.18×10^{-18} J (1 − 1/36)]

(C) [2.18×10^{-18} J (1 − 1/36)] / hc

(D) hc / [2.18×10^{-18} J (1/36)]

_____ **16.** An atom has only four energy levels: $E_1 = -9.0\times10^{-19}$ J, $E_2 = -4.0\times10^{-19}$ J, $E_3 = -2.0\times10^{-19}$ J, $E_4 = -1.0\times10^{-19}$ J. Which photon energy could appear in this atom's emission spectrum?

(A) 3.0×10^{-19} J

(B) 4.0×10^{-19} J

(C) 6.0×10^{-19} J

(D) 9.0×10^{-19} J

_____ **17.** Which statement is part of Bohr's model of the hydrogen atom?

(A) The electron gives off energy continuously as it orbits the nucleus.

(B) The electron can orbit at any distance from the nucleus.

(C) An electron in an allowed orbit has a fixed energy and does not radiate energy.

(D) The exact position and exact speed of the electron cannot both be known.

_____ **18.** Which statement about a hydrogen atom in its ground state is correct?

(A) Its electron is in n = 1, the lowest-energy and most stable level.

(B) Its electron has an energy of zero.

(C) It can emit a photon by moving its electron to a lower level.

(D) Its electron is in the n = 0 level.

_____ **19.** Which set of quantum numbers (n, l, m_l, m_s) is NOT allowed?

(A) (3, 2, −2, −½)

(B) (2, 1, 2, +½)

(C) (4, 2, 0, +½)

(D) (1, 0, 0, −½)

_____ **20.** Which subshell designation is NOT possible?

(A) 2p

(B) 3d

(C) 3f

(D) 6s

_____ **21.** How many orbitals are in the n = 3 shell, and what is the maximum number of electrons it can hold?

(A) 9 orbitals; 18 electrons
(B) 18 orbitals; 36 electrons
(C) 3 orbitals; 6 electrons
(D) 5 orbitals; 10 electrons

_____ **22.** A subshell has l = 1. What is the shape of its orbitals, and how many orbitals are in the subshell?

(A) Sphere; 1 orbital
(B) Cloverleaf; 5 orbitals
(C) Dumbbell; 6 orbitals
(D) Dumbbell; 3 orbitals

_____ **23.** How do the energies of the 3s, 3p and 3d orbitals compare in a hydrogen atom and in a sodium atom?

(A) H: 3s < 3p < 3d. Na: 3s < 3p < 3d.
(B) H: 3s = 3p = 3d. Na: 3s < 3p < 3d.
(C) H: 3s < 3p < 3d. Na: 3s = 3p = 3d.
(D) H: 3s = 3p = 3d. Na: 3s = 3p = 3d.

_____ **24.** A student draws oxygen's ground-state 2p electrons as [↑↓] [↑↓] [ ]. Which rule does this diagram violate?

(A) Pauli exclusion principle
(B) Aufbau principle
(C) Hund's rule
(D) Uncertainty principle

_____ **25.** Which orbital diagram violates the Pauli exclusion principle?

(A) 2s: [↑↑]
(B) 2s: [↑↓]
(C) 2p: [↑↓] [↑] [↑]
(D) 2p: [↑] [↑] [↑]

_____ **26.** What is the ground-state electron configuration of iodine (I)?

(A) [Kr] $5s^2\ 5p^5$
(B) [Kr] $5s^2\ 4d^5$
(C) [Kr] $5s^2\ 4d^{10}\ 5p^6$
(D) [Kr] $5s^2\ 4d^{10}\ 5p^5$

_____ **27.** What is the ground-state electron configuration of V^{3+}?

(A) [Ar] $4s^2$
(B) [Ar] $3d^2$
(C) [Ar] $3d^3$
(D) [Ar] $4s^1\ 3d^1$

_____ **28.** Which ground-state atom has exactly three unpaired electrons?

(A) Al
(B) S
(C) P
(D) Cl

_____ **29.** Which electron configuration represents an atom in an EXCITED state?

(A) $1s^2\ 2s^2\ 2p^6\ 3s^2$

(B) [Ar] $4s^2\ 3d^1$

(C) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

(D) $1s^2\ 2s^2\ 2p^6\ 3s^1\ 3p^1$

_____ **30.** Using Slater's rules, what is Zeff for a 3p electron in a phosphorus atom?

(A) 3.9

(B) 4.8

(C) 5.0

(D) 6.2

_____ **31.** Atomic radius decreases from Na to Cl across period 3. Which best explains this trend?

(A) Zeff on the valence electrons increases: each element adds a proton, but the added electron goes into the same shell and shields poorly.

(B) Each element adds its new electron to a higher energy level (larger n).

(C) The number of core electrons increases across the period, so shielding increases.

(D) More electrons means more electron–electron repulsion, which pulls the electron cloud inward.

_____ **32.** Which lists the atoms in order of INCREASING atomic radius?

(A) Sr < Mg < P < Cl

(B) Mg < P < Cl < Sr

(C) Cl < P < Mg < Sr

(D) P < Cl < Mg < Sr

_____ **33.** Which correctly compares the radii?

(A) $Li^+ > Li$ and $O^{2-} > O$

(B) $Li > Li^+$ and $O > O^{2-}$

(C) $Li^+ > Li$ and $O > O^{2-}$

(D) $Li > Li^+$ and $O^{2-} > O$

_____ **34.** Se^{2-}, Br^-, Rb^+ and Sr^{2+} are isoelectronic. Which of the following has the smallest radius?

(A) Sr^{2+}

(B) Rb^+

(C) Se^{2-}

(D) All are the same size, because they have the same number of electrons.

_____ **35.** The Br–Br bond length in Br_2 is 2.28 Å, and a C–C single bond is 1.54 Å. Using bonding atomic radii, estimate the C–Br bond length.

(A) 0.74 Å

(B) 1.91 Å

(C) 2.28 Å

(D) 3.82 Å

_____ **36.** The first five ionization energies (kJ/mol) of a period 3 element are 786, 1577, 3232, 4356 and 16,091. Which element is it?

(A) Al
(B) Si
(C) P
(D) S

_____ **37.** Which atom has the largest SECOND ionization energy?

(A) K
(B) Ca
(C) Ga
(D) Ge

_____ **38.** Which best explains why helium has the highest first ionization energy of all the elements?

(A) Helium is the most electronegative element, so it holds its electrons most tightly.
(B) Helium has the most protons of any noble gas.
(C) Its electrons are in n = 1, very close to the nucleus, and no core electrons shield them.
(D) Its electrons are farther from the nucleus than in any other atom.

_____ **39.** Which equation represents the electron affinity of sulfur?

(A) $S(g) \rightarrow S^{+}(g) + e^{-}$
(B) $S(g) + 2e^{-} \rightarrow S^{2-}(g)$
(C) $S^{-}(g) \rightarrow S(g) + e^{-}$
(D) $S(g) + e^{-} \rightarrow S^{-}(g)$

_____ **40.** Which best explains why cesium forms Cs^{+} ions so easily?

(A) Cs has a high first ionization energy, which makes the Cs^{+} ion very stable.
(B) Cs is highly electronegative, so it pulls electrons away from other atoms.
(C) Cs has a very low first ionization energy because its single 6s electron is far from the nucleus and well shielded.
(D) Losing an electron makes the Cs^{+} ion larger than the Cs atom.

Section II: Free Response

Directions: Show all work. Write the formula first, plug in values with units, and report answers with units and correct significant figures.

Question 1: Light and the hydrogen atom (10 points)

A blue laser emits light with a wavelength of 473 nm. Parts (c) through (e) are about the hydrogen atom. No constants are provided. For each calculation, write the formula, show the substitution with units, and give the answer with correct units and significant figures.

(a) Calculate the frequency of the laser light. (2 points)

(b) Calculate the energy of one photon of this light. The laser then fires a pulse with a total energy of 2.50 mJ; calculate the number of photons in the pulse. (2 points)

(c) An electron in a hydrogen atom moves from n = 5 to n = 3. Calculate ΔE for the electron, then calculate the wavelength, in nm, of the photon involved. (2 points)

(d) Is a photon emitted or absorbed in the transition in part (c)? Justify your answer using the sign of ΔE. Then identify the region of the electromagnetic spectrum that this photon belongs to. (2 points)

(e) A hydrogen atom has its electron in n = 2. Calculate the energy needed for the n = 2 → n = 4 and n = 2 → n = 5 transitions, and use these values to decide whether the atom can absorb one photon of the 473 nm laser light from part (b). Then explain why hydrogen produces a line spectrum instead of a continuous spectrum. (2 points)

Question 2: Ionization energies, Zeff and ion size (10 points)

Element X is in period 4. Its first four ionization energies are IE_1 = 590 kJ/mol, IE_2 = 1145 kJ/mol, IE_3 = 4912 kJ/mol and IE_4 = 6491 kJ/mol. A periodic table may be used. Show all work and justify answers in complete sentences.

(a) Identify element X. Justify your answer using the data. (2 points)

(b) Write the complete ground-state electron configuration of X, and the condensed electron configuration of X^{2+}. (2 points)

(c) Use Slater's rules to calculate Zeff for a 4s electron in X. Show how you found the shielding constant S. (2 points)

(d) Slater's rules give a 5s electron in strontium (Sr) the same Zeff as the 4s electron of X in part (c). Yet IE_1 of X is greater than IE_1 of Sr (549 kJ/mol). Explain this difference. (2 points)

(e) X^{2+} and Cl^- are isoelectronic. Use Slater's rules to calculate Zeff for a 3p electron in each ion, then use these values to justify which ion has the larger radius. (2 points)

Periodic Table of the Elements

1 **H** 1.01																	2 **He** 4.00
3 **Li** 6.94	4 **Be** 9.01											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
11 **Na** 22.99	12 **Mg** 24.30											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.06	17 **Cl** 35.45	18 **Ar** 39.95
19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.87	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.84	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.38	31 **Ga** 69.72	32 **Ge** 72.63	33 **As** 74.92	34 **Se** 78.97	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.95	43 **Tc** (98)	44 **Ru** 101.07	45 **Rh** 102.91	46 **Pd** 106.42	47 **Ag** 107.87	48 **Cd** 112.41	49 **In** 114.82	50 **Sn** 118.71	51 **Sb** 121.76	52 **Te** 127.60	53 **I** 126.90	54 **Xe** 131.29
55 **Cs** 132.91	56 **Ba** 137.33	57 **La** 138.91	72 **Hf** 178.49	73 **Ta** 180.95	74 **W** 183.84	75 **Re** 186.21	76 **Os** 190.23	77 **Ir** 192.22	78 **Pt** 195.08	79 **Au** 196.97	80 **Hg** 200.59	81 **Tl** 204.38	82 **Pb** 207.20	83 **Bi** 208.98	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 **Ac** (227)	104 **Rf** (261)	105 **Db** (262)	106 **Sg** (266)	107 **Bh** (264)	108 **Hs** (277)	109 **Mt** (268)	110 **Ds** (281)	111 **Rg** (272)	112 **Cn** (285)	113 **Nh** (286)	114 **Fl** (289)	115 **Mc** (289)	116 **Lv** (293)	117 **Ts** (294)	118 **Og** (294)

*Lanthanides	58 **Ce** 140.12	59 **Pr** 140.91	60 **Nd** 144.24	61 **Pm** (145)	62 **Sm** 150.36	63 **Eu** 151.96	64 **Gd** 157.25	65 **Tb** 158.93	66 **Dy** 162.50	67 **Ho** 164.93	68 **Er** 167.26	69 **Tm** 168.93	70 **Yb** 173.04	71 **Lu** 174.97
†Actinides	90 **Th** 232.04	91 **Pa** 231.04	92 **U** 238.03	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (262)

Atomic number above the symbol; average atomic mass below. Masses in parentheses are for the most stable isotope.

Answer Key: Practice Test B

Section I answers

1. B	**2.** D	**3.** A	**4.** D	**5.** B	**6.** C	**7.** C	**8.** D
9. A	**10.** B	**11.** D	**12.** A	**13.** C	**14.** D	**15.** B	**16.** A
17. C	**18.** A	**19.** B	**20.** C	**21.** A	**22.** D	**23.** B	**24.** C
25. A	**26.** D	**27.** B	**28.** C	**29.** D	**30.** B	**31.** A	**32.** C
33. D	**34.** A	**35.** B	**36.** B	**37.** A	**38.** C	**39.** D	**40.** C

Why each answer is right (and the tempting wrong one)

1. B Frequency goes up as wavelength goes down, so the shortest-wavelength radiation (gamma rays) has the highest frequency and radio the lowest. (A) is the order of decreasing WAVELENGTH, the exact reverse.

2. D $\nu = c/\lambda = (3.00\times10^{8}\ \text{m/s})/(7.50\times10^{-7}\ \text{m}) = 4.00\times10^{14}\ \text{s}^{-1}$, after converting 750 nm to meters. (C) divides by 750 without converting nm to m.

3. A 100 MHz = $1.00\times10^{8}\ \text{s}^{-1}$, so $\lambda = c/\nu = (3.00\times10^{8}\ \text{m/s})/(1.00\times10^{8}\ \text{s}^{-1}) = 3.00$ m. (C) treats 100 MHz as 100 Hz, forgetting that M = 10^{6}.

4. D Rearranging $E = hc/\lambda$ gives $\lambda = hc/E$ in meters (about 5.0×10^{-7} m), and multiplying by 10^{9} nm/m converts to nanometers (about 497 nm). (B) converts the wrong way: multiplying meters by 10^{-9} gives a meaningless tiny number, not nm.

5. B $E = hc/\lambda$, so photon energy is inversely proportional to wavelength: three times the wavelength gives one-third the energy. (A) gets the direction backward; a longer wavelength means a lower-energy photon.

6. C One photon: $E = hc/\lambda \approx (2.0\times10^{-25}\ \text{J·m})/(2.0\times10^{-7}\ \text{m}) \approx 1.0\times10^{-18}$ J; times 6.0×10^{23} photons/mol ≈ 6.0×10^{5} J/mol = 6.0×10^{2} kJ/mol. (D) is the right number in J/mol but skips dividing by 1000 to get kJ.

7. C A bigger drop releases more energy, which means a shorter wavelength: 2→1 is the smallest drop (121.5 nm), 3→1 is next (102.5 nm), then 4→1 (97.2 nm) and 5→1 (94.9 nm). (B) comes from reversing the order, as if a bigger drop gave a longer wavelength.

8. D Each photon must carry at least the threshold energy, and $E = h\nu$, so the fix is higher-frequency (shorter-wavelength) light such as UV. (A) is the trap: brighter light just sends more photons that each still have too little energy, so no electrons come off.

9. A The photon has more than the minimum energy, so an electron is ejected and the extra energy becomes kinetic energy: $5.0\times10^{-19} - 3.2\times10^{-19} = 1.8\times10^{-19}$ J. (C) adds the two energies instead of subtracting the energy used up freeing the electron.

10. B Energy levels in the atom are quantized, so each emitted photon has an energy (and wavelength, $E = hc/\lambda$) equal to the difference between two allowed levels, giving separate lines. (A) is false: hydrogen also emits ultraviolet and infrared lines, and even its visible light is a few lines, not a full rainbow.

11. D Dropping from n = 6 to n = 4 lowers the electron's energy: $\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/4^{2} - 1/6^{2})$ is negative, and the lost energy leaves as a photon. (B) has the right process but the wrong sign; emission always means $\Delta E < 0$.

12. A $E_n = -2.18\times10^{-18}\ \text{J}\ (1/n^{2})$ gets closer to zero as n increases, and the gaps shrink (the n = 1→2 gap is far bigger than n = 3→4). (C) is wrong because the $1/n^{2}$ term makes the levels crowd together, not stay evenly spaced.

13. C $E_3 = -2.18\times10^{-18}\ \text{J}\ (1/3^{2}) = -2.18\times10^{-18}\ \text{J} \div 9 \approx -2.42\times10^{-19}$ J. (D) drops the negative sign; every allowed level of a bound electron has negative energy, and only n = ∞ is zero.

14. D Highest frequency means the largest energy drop: 4→1 has (1 − 1/16), bigger than 2→1 (1 − 1/4) or 6→2 (1/4 − 1/36). (A) has an even bigger gap, but moving UP to n = 5 absorbs a photon; it doesn't emit one.

15. B The photon's energy is the size of ΔE: $2.18\times10^{-18}\ \text{J}\ (1/1^{2} - 1/6^{2}) = 2.18\times10^{-18}\ \text{J}\ (1 - 1/36)$, and $\lambda = hc/E$. (A) forgets to square n, using 1/6 instead of 1/36.

16. A An emitted photon's energy must equal the gap between two levels: $E_4 - E_2 = -1.0 - (-4.0) = 3.0\times10^{-19}$ J (the only possible gaps are 1.0, 2.0, 3.0, 5.0, 7.0 and 8.0×10^{-19} J). (B) is just the energy of level 2, not a difference between two levels.

17. C Bohr said the electron moves only in allowed orbits (specific n), each with a fixed energy, and it does not radiate while in one; light is emitted or absorbed only when it jumps between orbits. (A) is the classical-physics idea Bohr rejected.

18. A The ground state is n = 1, the lowest (most negative) energy and the most stable level. (C) is impossible: there is no level below n = 1, so a ground-state atom can only absorb energy, not emit it.

19. B m_l must run from −l to +l; with l = 1, m_l can only be −1, 0 or +1, so $m_l = 2$ breaks the rule. (A) looks odd with two negative values, but it is allowed: l = 2 is less than n = 3, $m_l = -2$ is within −2 to +2, and $m_s = -½$ is fine.

20. C For n = 3, l can only be 0, 1 or 2 (s, p, d), so an f subshell (l = 3) first appears at n = 4. (B) 3d does exist; it just fills after 4s, which is why it shows up in period 4.

21. A The n = 3 shell has 3s (1) + 3p (3) + 3d (5) = 9 = n^2 orbitals, and at 2 electrons per orbital it holds $2n^2 = 18$ electrons. (B) uses $2n^2$ for the orbitals; $2n^2$ counts electrons, not orbitals.

22. D l = 1 is a p subshell: dumbbell-shaped orbitals with $m_l = -1, 0, +1$, so 3 orbitals. (C) gives 6, which is the number of electrons a p subshell holds, not the number of orbitals.

23. B In one-electron hydrogen, orbital energy depends only on n, so 3s = 3p = 3d; in many-electron Na, the s electron penetrates closer to the nucleus and feels a higher Zeff, so 3s < 3p < 3d. (A) wrongly applies the many-electron order to hydrogen.

24. C Hund's rule says electrons go one per orbital (same spin) before any pair up, so oxygen's four 2p electrons should be [↑↓] [↑] [↑]. (A) isn't broken: each paired orbital has one up and one down spin.

25. A Two electrons in the same orbital already share n, l and m_l, so they must have opposite spins; [↑↑] gives both the same four quantum numbers. (D) is fine: those electrons are in different p orbitals (different m_l), so having the same spin is allowed and is what Hund's rule calls for.

26. D I (Z = 53) = [Kr] (36) + $5s^2$ + $4d^{10}$ + $5p^5$ = 53 electrons. (A) skips the $4d^{10}$ that fills between 5s and 5p in period 5, so it only accounts for 43 electrons.

27. B V is [Ar] $4s^2$ $3d^3$, and transition metals lose their 4s electrons first: removing $4s^2$ and then one 3d electron gives [Ar] $3d^2$. (A) removes the 3d electrons first, the classic mistake.

28. C P is [Ne] $3s^2$ $3p^3$; Hund's rule puts the three 3p electrons in three separate orbitals, giving 3 unpaired. (A) Al has 3 valence electrons, but its two 3s electrons are paired, so only its single 3p electron is unpaired.

29. D $1s^2$ $2s^2$ $2p^6$ $3s^1$ $3p^1$ has magnesium's 12 electrons, but one has moved up to 3p before 3s is full, so it is an excited state. (B) is scandium's normal ground state, since 4s fills before 3d.

30. B For a 3p electron of P (Z = 15): S = 4 other n = 3 electrons × 0.35 + 8 n = 2 electrons × 0.85 + 2 n = 1 electrons × 1.00 = 1.40 + 6.80 + 2.00 = 10.20, so Zeff = 15 − 10.20 = 4.8. (D) leaves out the 0.35 shielding from the other four n = 3 electrons; 5.0 is the simple core = 1, valence = 0 estimate, not Slater's rules.

31. A Across a period the nuclear charge rises while the new electrons enter the same shell and shield each other poorly, so Zeff rises and pulls the valence shell in. (C) is wrong: Na through Cl all have the same [Ne] core, so core shielding doesn't change across the period.

32. C Radius grows down a group and toward the left of a period: Cl < P < Mg across period 3, and Sr (period 5, same group as Mg) is largest. (A) is the exact reverse, the result of flipping the trend arrows.

33. D Li^+ has lost its only 2s electron, leaving just the smaller n = 1 shell, so it is smaller than Li; O^{2-} has two extra electrons (more e^-–e^- repulsion, same nuclear charge), so it is larger than O. (B) wrongly assumes every ion is smaller than its atom; anions are larger.

34. A All have 36 electrons, so the ion with the most protons (Sr, Z = 38) has the highest Zeff and pulls its electrons in tightest. (C) is actually the largest: it has the fewest protons (34) holding the same 36 electrons.

35. B A bonding radius is half the distance between two bonded identical atoms: r(Br) = 2.28/2 = 1.14 Å and r(C) = 1.54/2 = 0.77 Å, so C–Br ≈ 1.14 + 0.77 = 1.91 Å. (D) adds the two full bond lengths instead of the two radii.

36. B The big jump comes after IE_4 (4356 → 16,091 kJ/mol), so the atom has 4 valence electrons and the 5th must come from the core: group 14 in period 3 is Si. (C) assumes five listed values means five valence electrons; P's jump would come after IE_5.

37. A K ([Ar] $4s^1$) loses its only valence electron first, so its second electron must come from the n = 3 core, much closer to the nucleus with a much larger Zeff. (D) has the highest FIRST ionization energy of the four, but its second electron is still a valence 4p electron.

38. C Helium's two electrons sit in the n = 1 shell right next to the nucleus with no inner electrons shielding them, so removing one takes more energy than for any other atom. (A) mixes up two records: fluorine, not helium, is the most electronegative element.

39. D Electron affinity is the energy change when ONE electron is added to a gaseous atom: $S(g) + e^- \rightarrow S^-(g)$. (B) adds two electrons at once; sulfur forms S^{2-} in compounds, but electron affinity is defined for a single electron.

40. C Metals have low ionization energies, and Cs has one of the lowest: its lone 6s electron is far from the nucleus and heavily shielded, so it is easily removed to form a cation. (A) is backward: a HIGH ionization energy would make forming a cation harder.

Section II: Worked Answers

Question 1: Light and the hydrogen atom

(a) Calculate the frequency of the laser light. (2)

Formula: $c = \lambda\nu$, so $\nu = c/\lambda$

Convert: $\lambda = 473\ \text{nm} \times (1\ \text{m} / 10^{9}\ \text{nm}) = 4.73\times10^{-7}\ \text{m}$

Substitute: $\nu = (2.998\times10^{8}\ \text{m/s}) / (4.73\times10^{-7}\ \text{m})$

Answer: $\nu = 6.34\times10^{14}\ \text{s}^{-1}$ (3 sig figs, set by 473 nm)

Scoring: 1 pt formula and substitution with λ in meters; 1 pt answer with units (s^{-1} or Hz) and 3 sig figs.

(b) Calculate the energy of one photon of this light. The laser then fires a pulse with a total energy of 2.50 mJ; calculate the number of photons in the pulse. (2)

Formula: $E = h\nu$ (same as $E = hc/\lambda$)

Substitute: $E = (6.626\times10^{-34}\ \text{J·s})(6.338\times10^{14}\ \text{s}^{-1})$

Answer: $E = 4.20\times10^{-19}\ \text{J}$ per photon

Formula: number of photons = total energy ÷ energy per photon

Convert: $2.50\ \text{mJ} \times (1\ \text{J} / 1000\ \text{mJ}) = 2.50\times10^{-3}\ \text{J}$

Substitute: $(2.50\times10^{-3}\ \text{J}) / (4.20\times10^{-19}\ \text{J/photon})$

Answer: 5.95×10^{15} photons (3 sig figs)

Scoring: 1 pt photon energy with formula, units and 3 sig figs; 1 pt number of photons with mJ converted to J.

(c) An electron in a hydrogen atom moves from n = 5 to n = 3. Calculate ΔE for the electron, then calculate the wavelength, in nm, of the photon involved. (2)

Formula: $\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/n_f^2 - 1/n_i^2)$

Substitute: $\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/3^2 - 1/5^2) = -2.18\times10^{-18}\ \text{J}\ (0.1111 - 0.0400)$

Answer: $\Delta E = -1.55\times10^{-19}\ \text{J}$

Formula: photon energy = $|\Delta E|$, and $\lambda = hc/E$

Substitute: $\lambda = (6.626\times10^{-34}\ \text{J·s})(2.998\times10^{8}\ \text{m/s}) / (1.55\times10^{-19}\ \text{J}) = 1.28\times10^{-6}\ \text{m}$

Convert: $1.28\times10^{-6}\ \text{m} \times (10^{9}\ \text{nm} / 1\ \text{m})$

Answer: $\lambda = 1.28\times10^{3}\ \text{nm}$

Scoring: 1 pt ΔE with the negative sign, units and 3 sig figs; 1 pt wavelength in nm with 3 sig figs.

(d) Is a photon emitted or absorbed in the transition in part (c)? Justify your answer using the sign of ΔE. Then identify the region of the electromagnetic spectrum that this photon belongs to. (2)

Emitted.

Justification: ΔE is negative ($n_f = 3$ is lower than $n_i = 5$), so the electron drops to a lower, more stable level and the energy it loses leaves the atom as a photon.

Region: 1.28×10^{3} nm is longer than visible light (about 400 to 750 nm), so the photon is infrared.

Scoring: 1 pt 'emitted' with the ΔE sign as the reason; 1 pt infrared, tied to the wavelength.

(e) A hydrogen atom has its electron in n = 2. Calculate the energy needed for the n = 2 → n = 4 and n = 2 → n = 5 transitions, and use these values to decide whether the atom can absorb one photon of the 473 nm laser light from part (b). Then explain why hydrogen produces a line spectrum instead of a continuous spectrum. (2)

Formula: $\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/n_f^2 - 1/n_i^2)$

$n = 2 \rightarrow 4$: $\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/4^2 - 1/2^2) = +4.09\times10^{-19}\ \text{J}$

$n = 2 \rightarrow 5$: $\Delta E = -2.18\times10^{-18}\ \text{J}\ (1/5^2 - 1/2^2) = +4.58\times10^{-19}\ \text{J}$

Decision: the 473 nm photon carries 4.20×10^{-19} J. That matches neither gap (it falls between them), so the atom cannot absorb it. (It is also less than the 5.45×10^{-19} J needed to remove the electron from n = 2.)

Explanation: the electron's energy is quantized, so an atom can only absorb or emit a photon whose energy exactly equals the difference between two allowed levels. Only certain energy differences exist, so only certain wavelengths ($\lambda = hc/\Delta E$) are emitted, which shows up as separate bright lines instead of a continuous spectrum.

Scoring: 1 pt both ΔE values and the 'cannot absorb' conclusion; 1 pt line-spectrum explanation tied to quantized energy levels.

Question 2: Ionization energies, Zeff and ion size

(a) Identify element X. Justify your answer using the data. (2)

X is calcium (Ca).

Justification: the large jump is between IE_2 (1145 kJ/mol) and IE_3 (4912 kJ/mol), more than 4 times larger, while IE_1 to IE_2 only about doubles. So X has 2 valence electrons (group 2). The third electron must come from the n = 3 core, which is closer to the nucleus and feels a much larger Zeff, so it takes far more energy to remove.

Group 2 in period 4 is Ca.

Scoring: 1 pt Ca; 1 pt justification that points to the $IE_2 \rightarrow IE_3$ jump and core electrons.

(b) Write the complete ground-state electron configuration of X, and the condensed electron configuration of X^{2+}. (2)

Ca: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

Ca^{2+}: [Ar] (the two 4s electrons are removed, leaving $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$)

Scoring: 1 pt each.

(c) Use Slater's rules to calculate Zeff for a 4s electron in X. Show how you found the shielding constant S. (2)

Formula: $Z_{eff} = Z - S$

Slater's rules: each other electron in the same n shields 0.35; each electron in n − 1 shields 0.85; each electron in n − 2 or lower shields 1.00.

Grouping for one 4s electron of Ca (Z = 20): 1 other electron in n = 4; 8 electrons in n = 3 ($3s^2\ 3p^6$); 10 electrons in n = 1 and 2.

Substitute: $S = 1(0.35) + 8(0.85) + 10(1.00) = 0.35 + 6.80 + 10.00 = 17.15$

Answer: $Z_{eff} = 20 - 17.15 = +2.85$

Scoring: 1 pt S with the electrons correctly grouped by shell; 1 pt Zeff = +2.85.

(d) Slater's rules give a 5s electron in strontium (Sr) the same Zeff as the 4s electron of X in part (c). Yet IE_1 of X is greater than IE_1 of Sr (549 kJ/mol). Explain this difference. (2)

Check: Sr 5s electron: S = 1(0.35) + 8(0.85) + 28(1.00) = 35.15, so Zeff = 38 − 35.15 = +2.85, the same as Ca.

Explanation: since Zeff is the same, the difference comes from distance. X's (Ca's) valence electron is in n = 4, while Sr's is in n = 5, so Ca's electron is closer to the nucleus and Ca has the smaller atomic radius.

The same Zeff attracts a closer electron more strongly, so more energy is needed to remove Ca's 4s electron: IE_1(Ca) > IE_1(Sr). This is why ionization energy decreases down a group.

Scoring: 1 pt Ca's valence electron is in a lower n (closer to the nucleus, smaller radius); 1 pt closer electron → stronger attraction → higher IE.

(e) X^{2+} and Cl^- are isoelectronic. Use Slater's rules to calculate Zeff for a 3p electron in each ion, then use these values to justify which ion has the larger radius. (2)

Both ions: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (18 electrons). Formula: Zeff = Z − S

S for a 3p electron = 7(0.35) + 8(0.85) + 2(1.00) = 2.45 + 6.80 + 2.00 = 11.25 (the same for both ions)

Cl^-: Zeff = 17 − 11.25 = +5.75

Ca^{2+}: Zeff = 20 − 11.25 = +8.75

Cl^- is larger. Both ions have the same electrons and the same shielding, but Cl^- has 3 fewer protons, so its 3p electrons feel a smaller Zeff (5.75 vs 8.75), are held less tightly and sit farther from the nucleus.

Scoring: 1 pt both Zeff values with S shown; 1 pt Cl^- larger, justified with Zeff.